1. Threat Hunting Practice

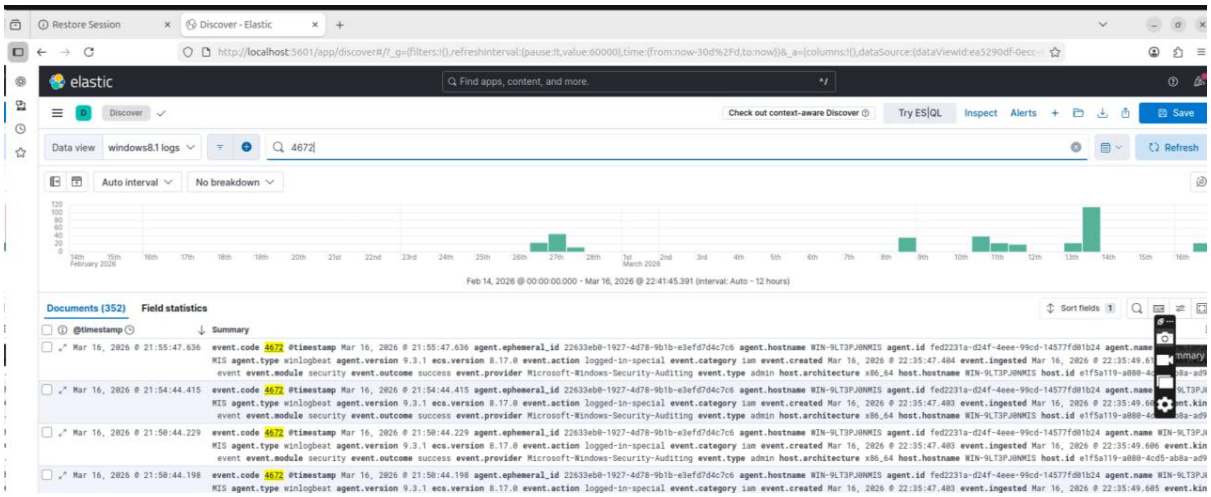
The threat hunting activity was conducted to proactively identify suspicious behavior related to unauthorized privilege escalation within the environment. The tools used for this activity include Elastic Security, Velociraptor, and AlienVault OTX.

Hypothesis Development:

A hunting hypothesis was formulated stating that “Unauthorized privilege escalation may have occurred in domain accounts.” To validate this, logs were analyzed in Elastic Security by filtering Event ID 4672, which indicates special privileges assigned to a new logon.

Log Analysis:

The following suspicious event was identified during log analysis:



Timestamp	User	Event ID	Notes
3/18/2026 15:22	TestUser	4672	Unexpected admin role assigned

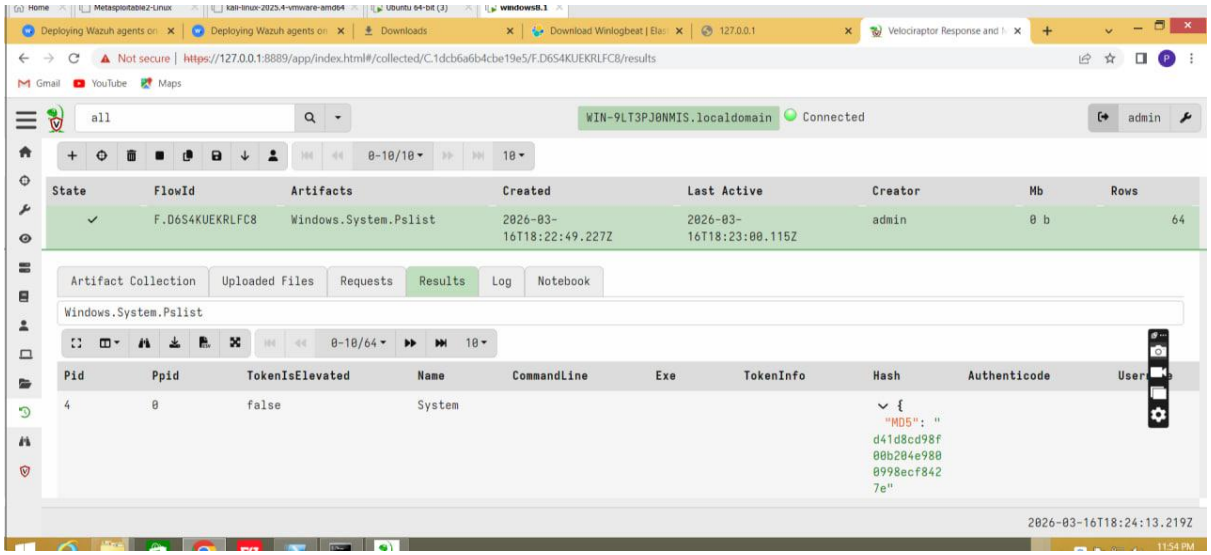
This event indicates that the user “testuser” was granted elevated privileges unexpectedly, which could be a sign of account compromise or misuse of administrative rights.

Threat Intelligence Correlation:

To enrich the investigation, threat intelligence data was gathered using AlienVault OTX. Indicators of Compromise (IOCs) related to MITRE ATT&CK technique T1078 (Valid Accounts) were analyzed. Suspicious IP addresses and patterns were identified and cross-referenced with endpoint data using Velociraptor queries:

```
SELECT * FROM processes
```

The process analysis helped identify abnormal or unauthorized processes running under privileged accounts.

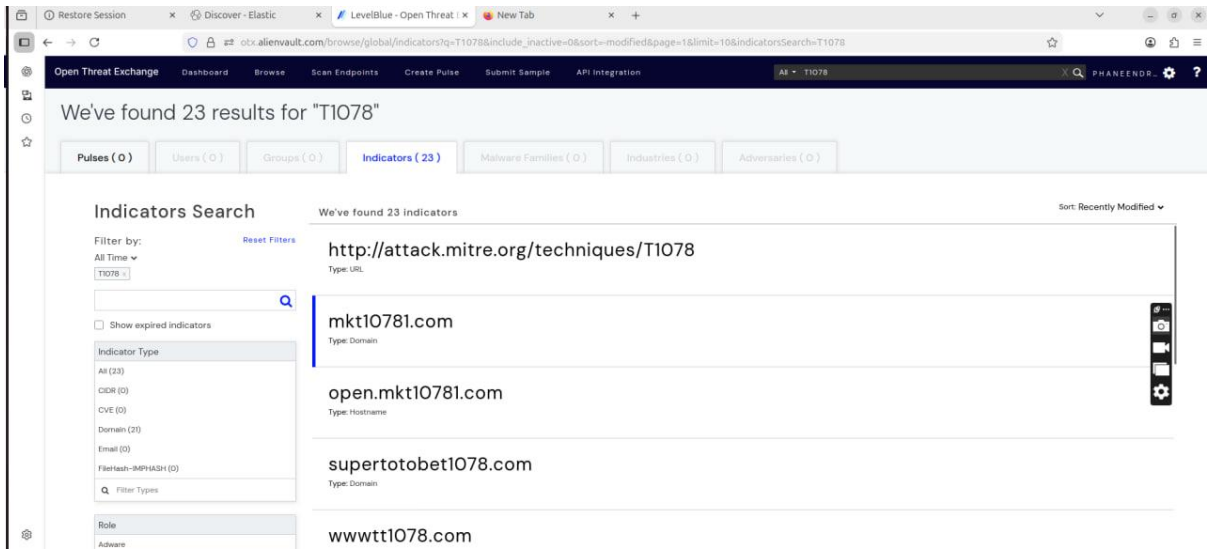


Findings:

The investigation revealed that privileged access was granted without proper authorization, and correlated threat intelligence suggested possible misuse of valid credentials. No direct malware was identified, but the activity indicates potential insider threat or credential compromise.

MITRE ATT&CK Mapping:

T1078 – Valid Accounts



Hunting Report (100 Words):

A proactive threat hunting activity was conducted to identify unauthorized privilege escalation within the system. Using Elastic Security, Event ID 4672 logs were analyzed, revealing suspicious privilege assignment to a user account. Threat intelligence from AlienVault OTX was used to identify related indicators of compromise, which were cross-validated using Velociraptor process queries. The findings suggest potential misuse of valid credentials, aligning with MITRE ATT&CK technique T1078. Although no malicious payloads were detected, the activity indicates a possible security risk. It is recommended to enforce strict access controls, monitor privileged accounts, and implement multi-factor authentication to prevent unauthorized access.